Introduction to Accelerated Genomics

Applied Machine Learning for Biological Data - Module 2, Day 5

- Date: June 6, 2025
- Time: 09:00-16:00

⚙ Prerequisites

Follow BioNT course - A practical introduction to bioinformatics or equivalent knowledge

Learning outcomes

- Gain a foundational understanding of the NVIDIA Parabricks software suite
- Acquire practical experience in using Parabricks tools for basic germline variant calling workflow

Overall schedule

- Introduction to Docker
- Introduction to GPU
- Introduction to NGS
- Basic CPU-native NGS data processing workflow
- Introduction to NVIDIA Parabricks
- Parabricks Hands-on
- Parabricks vs CPU-native NGS data processing

Introduction to NGS

Next Generation Sequencing (NGS)

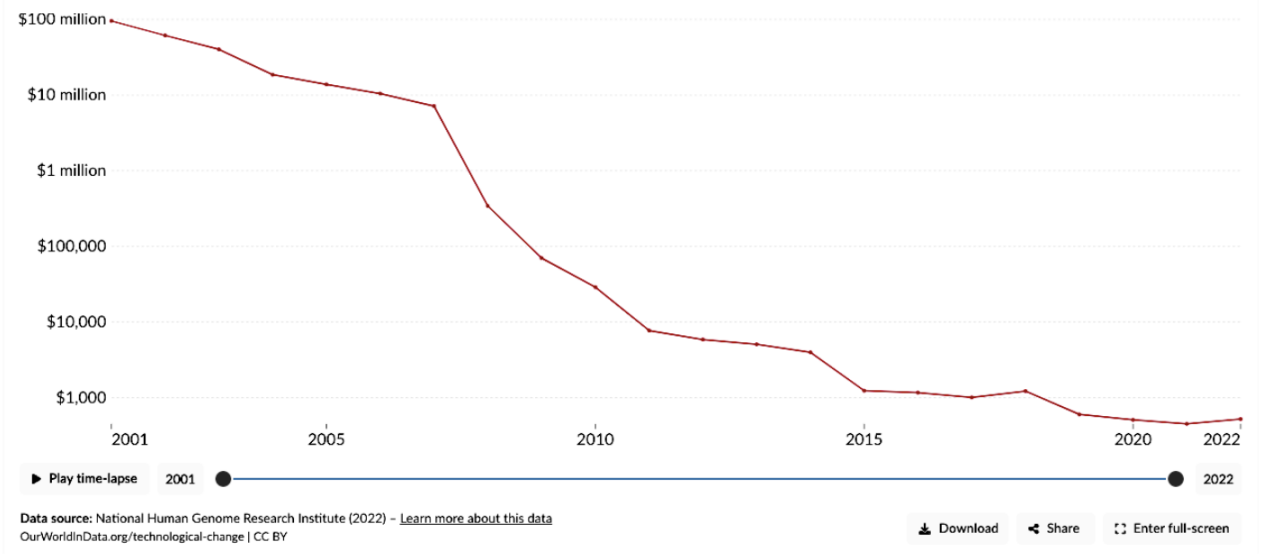
- DNA Sequencing process
  - Process of determining the order of nucleotides in a genome (genome codes the complete set of instructions for cellular functions)
- Next Generation Sequencing (NGS)
  - Increasing the throughput by massively parallelising the sequencing process

Sequencing throughput and sequencing cost

- Today, we are no longer limited by the availability of sequencing data due to the rapidly decreasing cost

Sequencing cost over the years

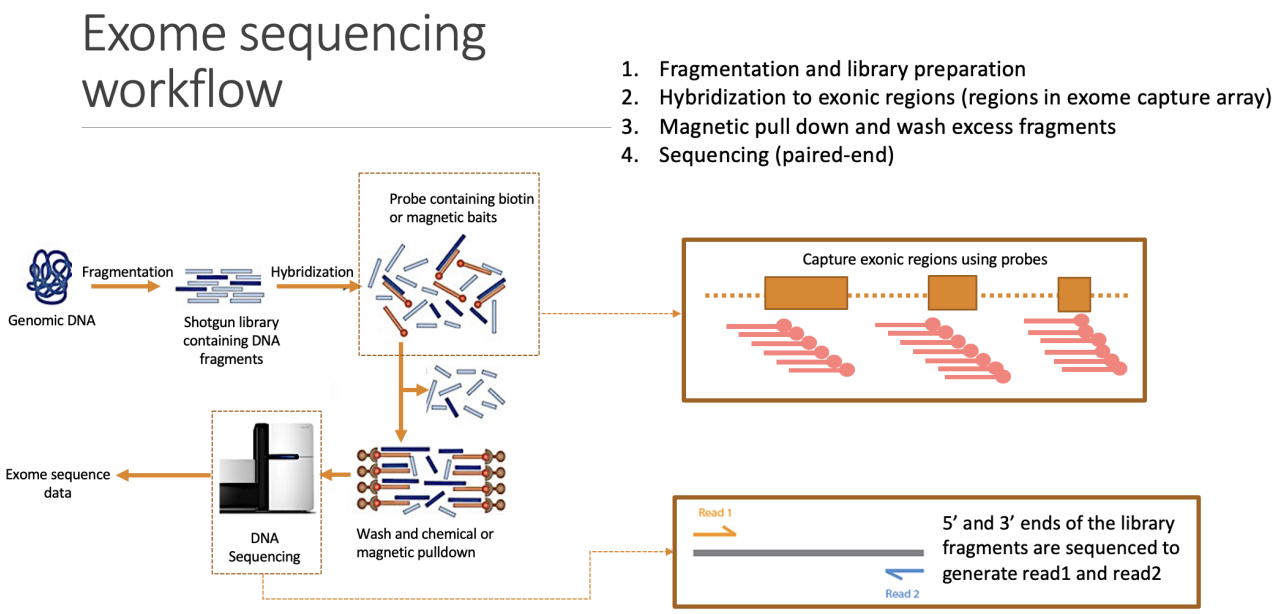
Sequencing throughput over the years



https://ourworldindata.org/grapher/cost-of-sequencing-a-full-human-genome

Sequencing tech used in the Human Genome Project

Main steps of DNA sequencing



NGS data analysis (Germline variant calling)

Main steps in NGS analysis

```
    graph TD;
i1(Raw data) --> 1
i2(Reference data) --> 1
1(Mapping raw data onto the reference)
1 -- Alignment file --> 2
2(Optimization & fine-tuning of the alignment file)
2 -- Optimised & fine-tuned alignment file --> 3
3(Identify DNA changes)

classDef highlight fill:#99ccff;
classDef white fill:#ffffff;
class 1,2,3, highlight;
```

Precision genomics

- Tailoring disease prevention and treatment based on the person's genetic makeup
- Here, we analyse raw sequencing data and identify changes in DNA that could lead to a diseases
- Raw data analysis involves a number of complex processes and the collections of such processes are commonly known as "Sequence analysis workflows"

Precision genomics in routine clinical practice

- Sequence analysis workflows are routinely used in current clinical setting
- However they all (a large majority) use CPUs for the computing power

Basic CPU-native NGS data processing workflow

- Simplest NGS data processing workflows have two main stages
  - Read mapping and alignment refinement
  - Variant calling
- **Test data:** `https://s3.amazonaws.com/parabricks.sample/parabricks_sample.tar.gz`
- **Source of the test data -** `NVIDIA tutorials` `<https://docs.nvidia.com/clara/parabricks/4.3.0/tutorials/gettingthesampledata.html>`_

Stage 1: Read mapping and alignment refinement

Step 1: BWA read mapping and sorting

```bash
#!/bin/bash

SAMPLE_NAME="pb_sample"

FASTA="Ref/Homo_sapiens_assembly38.fasta"
READ1="Data/sample_1.fq.gz"
READ2="Data/sample_2.fq.gz"
BAM_OUT="${SAMPLE_NAME}_CPU.bam"

## Run BWA MEM
### Source: https://github.com/nf-core/sarek/blob/f034b737630972e90aeae851e236f9d4292b9a4f/modules/nf-core/bwa/mem/main.nf#L30
bwa mem \
-t 64 \
-R "@RG\tID:rg1\tLB:lib1\tPL:bar\tSM:SM1\tPU:SM1_rg1" \
${FASTA} \
${READ1} ${READ2} \
| samtools     sort     -@64     -o ${BAM_OUT} -

samtools index -@64 -b ${BAM_OUT}
```

Step 2: GATK MarkDuplicates

```bash
#!/bin/bash

SAMPLE_NAME="pb_sample"
BAM_IN="${SAMPLE_NAME}_CPU.bam"
BAM_OUT="${SAMPLE_NAME}_markdup.bam"

## Mark duplicated reads in BAM file
gatk --java-options  "-Xss3m -XX:-UsePerfData" MarkDuplicates \
--VALIDATION_STRINGENCY SILENT \
-I ${BAM_IN} \
-O ${BAM_OUT} \
-M metrics.txt \
--TMP_DIR temp_dir

samtools index ${BAM_OUT}
```

Step 3: GATK BaseRecalibrator

```bash
#!/bin/bash

FASTA="Ref/Homo_sapiens_assembly38.fasta"
KNOWN_SITES="Ref/Homo_sapiens_assembly38.known_indels.vcf.gz"

SAMPLE_NAME="pb_sample"
BAM_IN="${SAMPLE_NAME}_markdup.bam"
RECAL_FILE="CPU_recal.txt"

## Generate BQSR Report
gatk --java-options  "-XX:-UsePerfData" BaseRecalibrator \
--input ${BAM_IN} \
--output ${RECAL_OUT} \
--known-sites ${KNOWN_SITES} \
--reference ${FASTA}
```

Step 4: GATK ApplyBQSR

```bash
#!/bin/bash

FASTA=GRCh38"/"GCA_000001405.15_GRCh38_no_alt_analysis_set.fna
BAM_IN="${SAMPLE_NAME}_markdup.bam"
RECAL_FILE="CPU_recal.txt"
BAM_OUT="${SAMPLE_NAME}_CPU_markdup_BQSR.bam"

## Run ApplyBQSR Step
gatk --java-options  "-XX:-UsePerfData" ApplyBQSR \
-R ${FASTA} \
-I ${BAM_IN} \
--bqsr-recal-file ${RECAL_FILE} \
-O ${BAM_OUT}
```

Stage 2: Variant calling via HaplotypeCaller

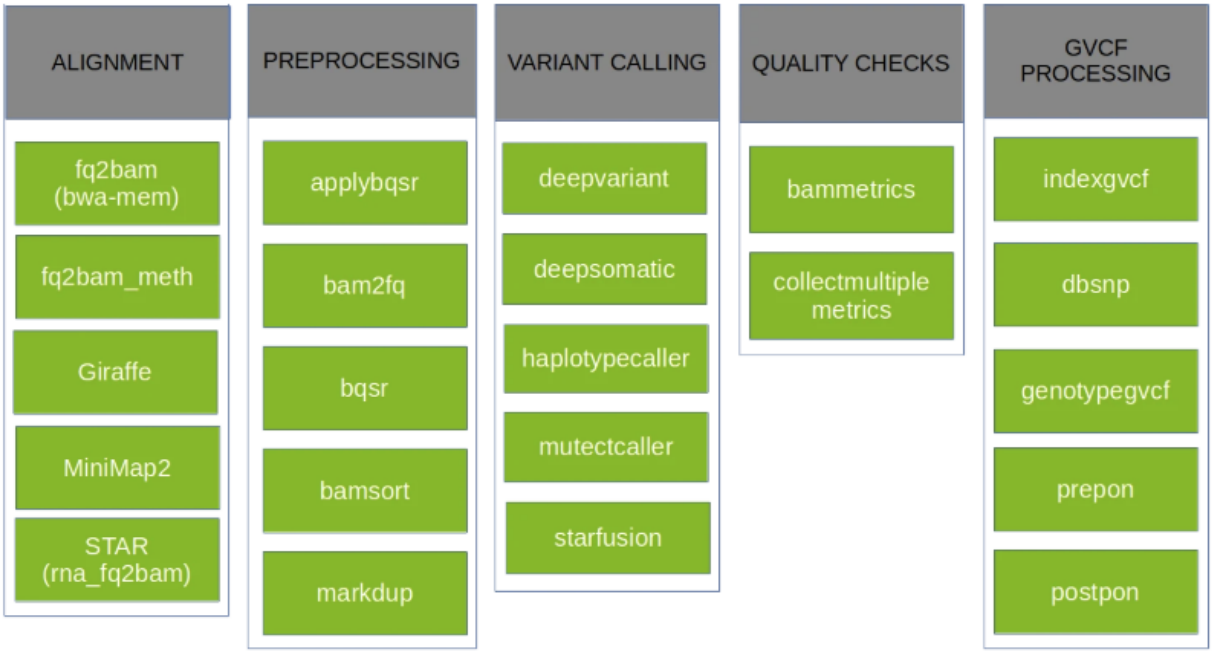
```bash
#!/bin/bash

set -euo pipefail

SAMPLE_NAME="pb_sample"
FASTA="Ref/Homo_sapiens_assembly38.fasta"
BAM_IN="${SAMPLE_NAME}_CPU_markdup_BQSR.bam"

## Variant calling with HaplotypeCaller
gatk --java-options  "-XX:-UsePerfData" HaplotypeCaller \
--input ${BAM_IN} \
--output ${BAM_IN}.vcf \
--reference ${FASTA} \
--native-pair-hmm-threads 64
```

NVIDIA Parabricks

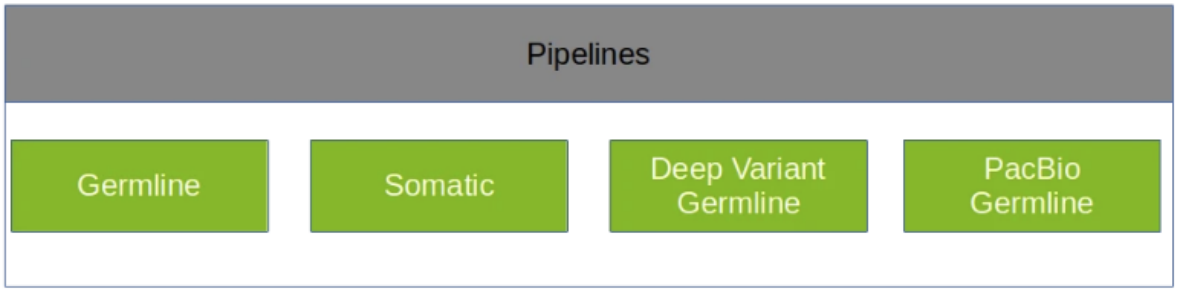
- Parabricks offers range of accelerated NGS data processing tools/commands
- Parabricks documentation:

- Parabricks was built from the ground up by GPU computing and Deep Learning experts who wanted to develop the fastest and most efficient possible implementation of common genomics algorithms.
- We can use these commands and create a simple NGS data processing workflows with two main stages
  - Read mapping and alignment refinement
  - Variant calling

Parabricks supports the tools shown below:



Additionally, the Parabricks tool suite provides a number of variant calling pipelines that are each a combination of several individual tools, combining into one tool what would otherwise be a multi-step process. These pipelines are:

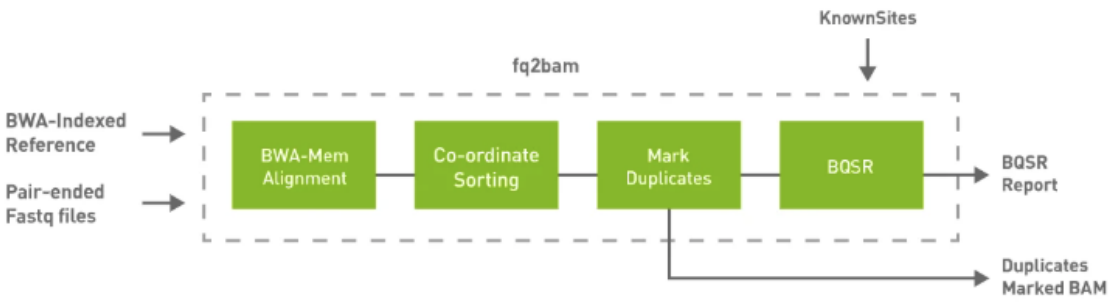


Ref:

Parabricks workflow (basic workflow)

Stage 1: Read mapping and alignment refinement

Step 1: fq2bam



Source

Step 2: applybqsr

- Compatible GATK4 Command: `gatk ApplyBQSR`

Stage 2: Variant calling

Parabricks HaplotypeCaller

Source

Hands-on session: Link

Results

Output files

```
-rw-r--r-- 1 root      root      4819386868 Dec  4 14:11 pbrun_fq2bam_GPU.bam
-rw-r--r-- 1 root      root         6882792 Dec  4 14:11 pbrun_fq2bam_GPU.bam.bai
-rw-r--r-- 1 root      root        22693986 Dec  4 14:26
pbrun_fq2bam_GPU.bam_applybqsr.vcf
-rw-r--r-- 1 root      root      4947983325 Dec  4 14:18 pbrun_fq2bam_GPU_applybqsr.bam
-rw-r--r-- 1 root      root         6882792 Dec  4 14:18
pbrun_fq2bam_GPU_applybqsr.bam.bai
-rw-r--r-- 1 root      root           87690 Dec  4 14:11 pbrun_fq2bam_GPU_chrs.txt
-rw-r--r-- 1 root      root          395623 Dec  4 14:11 pbrun_recal_gpu.txt
```

Variant counts

```
grep -vc "#" pbrun_fq2bam_GPU.bam_applybqsr.vcf
> 127720
```

Parabricks vs CPU-native NGS data processing

Runtime comparison and speedup calculation

- Speedup = Runtime of base-process / Runtime of accelerated-process

Step	Base-runtime	Accelerated-runtime	Speedup
Mapping and alignment refinement	1 h, 25 min, 14 sec	2 min, 39 sec	32.16
GATK ApplyBQSR	29 min, 36 sec	34 sec	52.24
GATK HaplotypeCaller	1 h, 35 min, 5 sec	3 min, 13 sec	29.55

Variant counts identified in CPU and Parabricks pipelines

NAIC accelerated genomics pipeline

- GitHub repository

Our current activities

Image sources: NVIDIA ; DRAGEN

- NAIC LinkedIn page